Roll No……………..

Mid Semester Examination 2025

Name of the Program: B.Tech(ECE) *Semester: VI*
Name of the Course: Wireless Communication *Course Code:* TEC 601

Time: 90 Minutes Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks

Q1	(10 marks)	**CO1**
(a)	(i) Explain meaning of the terms 'universal frequency reuse' and 'fractional frequency reuse'. (ii) Define dwell time and the factors on which the dwell time depends?	
OR		
(b)	Derive an expression to show the relationship between co-channel reuse ratio (Q) and the cluster size.	
Q2	(10 marks)	**CO1**
(a)	Assuming 7 cell cluster and UE at the cell boundary, find an expression to indicate SIR in co-channel cells in the downlink.	
OR		
(b)	Find SIR at the boundary of a cell in case of 7, 9, and 12 cells in a cluster. If the minimum required SIR for acceptable voice quality is 18 dB then which one cannot be preferred? Assume path loss exponent to be 4.	
Q3	(10 marks)	**CO1**
(a)	How cell splitting is performed? Why it can increase the system capacity?	
OR		
(b)	Explain Hard, Soft and Softer hand-offs.	
Q4	(10 marks)	**CO2**
(a)	Describe free space propagation model for mobile radio wave propagation.	
OR		
(b)	Consider a transmitter operating at 1800 MHz transmits 4 watt power. Assume path loss exponent to be 4, shadow effect of 10.5 dB and the power at reference point (d_0= 100m) is - 32 dB. Find the received power at a distance of 3 km from transmitter and the allowable path loss.	
Q5	(10 marks)	**CO2**
(a)	(i) Find the Fraunhofer distance for a transmitting antenna having the maximum dimension of 1m operating at 900 MHz frequency (ii) Define Fraunhofer region.	
OR		
(b)	Draw the block diagram of Wireless communication system and explain the Function of each block.	